

CO2 AT3

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1. An image appears to bright

Bitable grey level transformation: image
Negative or logically, Power-law transformation

Concept:

The Power law transformation modifies the intensity
Using the equation

$$S = C \times r^\gamma$$

r = input pixel intensity

S = output pixel intensity

γ = gamma value

C = constant

Expected outcome:

Overall brightness decreases

Hidden details become visible

Image appears more natural

High-intensity regions are preserved without
saturation.

2) A dark image lacks detail

Dark image contain most intensity values in the lower gray-level range. making detail invisible. the appropriate enhancement technique is Gamma correction

$$S = C \times r^{\gamma}$$

Application process :

1. read the image
2. choose gamma less than 1
3. Display the enhanced image

Expected outcome :

- Dark regions become brighter
- Hidden objects become visible
- Bright regions remain almost unchanged

3) An image has poor contrast

when an image has poor contrast, pixel intensities occupy only a small range. Histogram equalization spreads the intensity values across the full gray-level range.

Working Process :

1. Calculate histogram

2. Compute Cumulative histogram
 3. Normalize the CDF
 4. Generate enhanced image
- Expected Outcome :
- Contrast increases
 - image details become clearer
 - Edges become more visible

4) A low contrast image requires uniform intensity distribution

To achieve a uniform intensity distribution, histogram equalization is the most appropriate method because it redistributes pixel intensities across the gray-level range

Working Process :

1. Compute image histogram
2. Normalize cumulative values
3. Replace each pixel with its mapped intensity

Effect on intensity levels :

- Pixel values spread uniformly
- Dark regions become brighter
- Bright regions gain more detail
- Overall contrast improves significantly

5. An image shows uneven brightness across regions.

When different regions of an image have different brightness levels, contrast limited adaptive histogram equalization is the best enhancement technique.

Working Process:

1. Divide the image into small regions
2. Perform histogram equalization on each file
3. Merge neighboring tiles using interpolation

Expected Outcome:

- Corrects non-uniform illumination
- Improves local contrast
- Preserve image details
- Reduces noise amplification
- Produces a natural-looking enhanced image